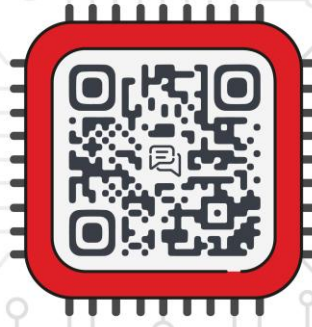




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كلية الهندسة المعلوماتية

السنة الثانية

Signal Analysis

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RB Informatics; 2026/5/18

الاتصالات الرقمية

Course content:

- Time Domain vs Frequency Domain
- Simple Signals vs Composite Signals
- Bandwidth
- Periodic vs Aperiodic Signals
- Fourier Analysis
- Newton prism
- Signal Decomposition
- Fourier Coefficients
- Odd and Even Signals
- Solved Examples
- Examples on Fourier Series and signal analysis

Time domain VS frequency domain

- The time domain is a method of analyzing physical signals, mathematical functions, or economic data by representing how they change over time.
- It shows the signal's amplitude or intensity on the y-axis against time on the x-axis, allowing for observation of **sequential** events, such as sound waves or voltage changes in an **Oscilloscope**.
- The frequency domain refers to the analysis of signals or mathematical functions with respect to frequency, rather than time. It shows how a **signal's energy** is distributed across various frequency bands, using components like magnitude and phase to represent the signal's frequency spectrum.

Explanation:

Time domain means that I am studying how the amplitude of a signal changes with time.

If we observe the professor's voice during the lecture, what will I get if I put it on the output of an **oscilloscope**?

I will get **oscillations**.

Because the amplitude is different and the frequency is different, I end up with multiple amplitudes and multiple frequencies.

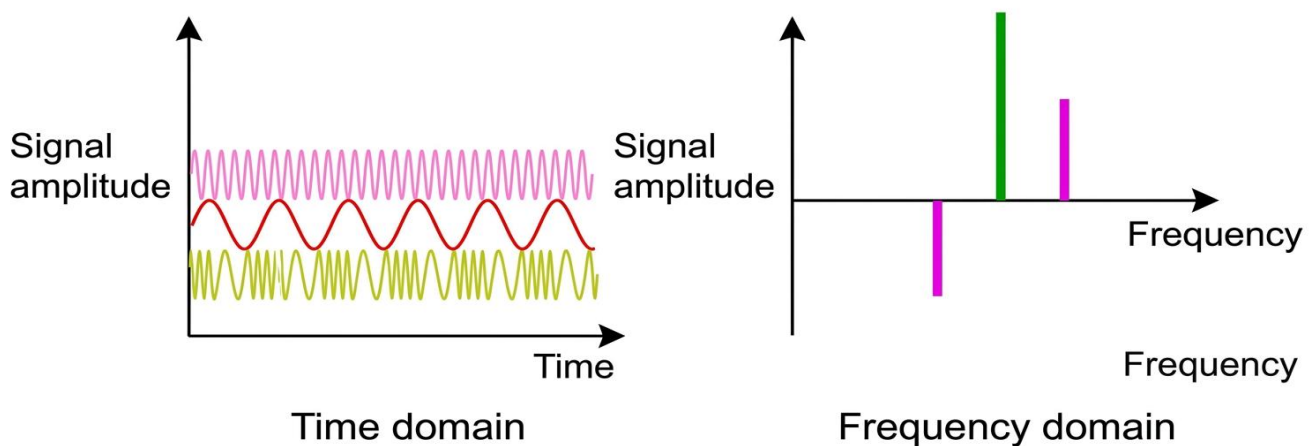
A **composite signal** cannot be fully studied in the time domain

In the **time domain**, we only analyze amplitude versus time. But a composite signal has multiple amplitudes, multiple frequencies, and multiple phases, so it becomes difficult to interpret it directly.

That is why we use another domain called the **frequency domain**.

The **frequency domain** shows how the energy of the signal is distributed across different frequency bands.

Instead of focusing on amplitude over time, we focus here on how the signal's energy is spread across frequencies.

Visual Comparison of Domains

The paradox of choice: More is less 🧑‍🔧

- **Time domain graph:** It shows three overlapping continuous sine waves of different frequencies
- **Frequency domain graph:** It displays the same information as three discrete vertical bars (impulses) at specific points on the frequency axis, indicating the strength of each individual frequency component.

In time domain, notice how many signals I have

I have three different signals. Each one has its own amplitude, each one has its own frequency, and each one has its own phase.

In the frequency domain, I deal only with frequency and amplitude, and I represent the values in a reversed form compared to the time domain.

This representation shows how the energy of the signal is distributed across different frequency bands

• Why Both Domains Matter?

Time domain limitation: When you have composite signals (professor's voice, music), the oscilloscope shows overlapping waves that are difficult to analyze.

Solution: Convert to frequency domain to see individual frequency components clearly as separate vertical bars.

Student Note: Time domain answers "What is happening NOW?" Frequency domain answers "What frequencies make up this signal?"

Simple VS Composite signals

Simple signals	Composite signals
Simple signals are single-frequency sine waves that cannot be decomposed further	composite signals are combinations of multiple simple sine waves with different frequencies, amplitudes, and phases.
Simple signals (e.g., a pure tone) represent a single, pure frequency,	composite signals (e.g., speech, music, data signals) carry complex, multi-frequency information.
Examples: U(t), r(t), sin(x), Alarm signals or audio	Examples: Vocal, Video or visual composite signals

Alarm example explanation:

For example, there is an **alarm**. Before that, there was an audio **signal** that alerts you that you are not wearing your seatbelt.

This signal is a **simple signal**.

Now it has become visual, where you can see it on the car dashboard indicating that you are not wearing your seatbelt

A **composite signal** is an electromagnetic wave composed of multiple simple sine waves (different frequencies, amplitudes, and phases) combined into one, allowing it to carry **complex** information.

Examples of complex information: Vocal, video...

The bandwidth: The **bandwidth** of a composite signal is determined by the difference between its highest and lowest frequency components.

Explanation:

I am talking about a **composite signal**, which means that it consists of frequency components.

One component is 10 kHz, another component is 50 kHz, and another one is 10 MHz.

All of these are components that exist within the same signal.

The **bandwidth** is the difference between the highest frequency present, which is 10 MHz, and the lowest frequency, such as 10 kHz.

For example, on public roads between cities, we notice bridges, and on each bridge there is a sign indicating the allowed height.

What does this mean?

It means that a loaded vehicle arrives, and if the load exceeds the height of the bridge, there are two possibilities:

Either the car pulls the bridge along with it while moving, which is impossible 😊, or the bridge blocks everything that is above its height.

The conclusion is that part of the load does not reach its destination yet.

The same process applies to signals

We say that **bandwidth** is not only related to the signal, but also to the transmission medium. The medium itself has bandwidth.

You cannot transmit frequencies that exceed the allowed range of the signal or the medium. If you do, those frequency components will not pass, and therefore the receiver will not receive the full transmitted signal.

It is exactly like the **bridge example**.

In communications, data from the source must reach the destination exactly as it is.

The bandwidth of the signal and the bandwidth of the medium must match at least.

If the medium bandwidth is larger, that **is fine**. But if the signal bandwidth is larger than the medium, that is **not acceptable**, because some components will be lost

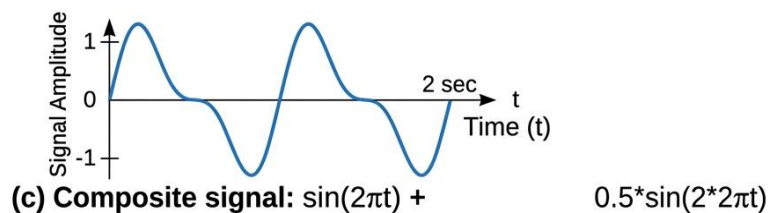
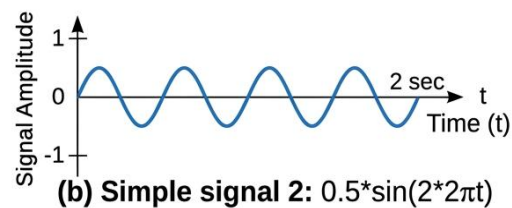
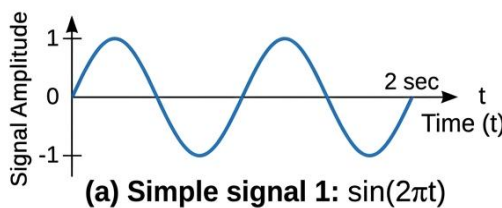
In each time period, the signal repeats

Composite signals

Periodic

vs

Aperiodic



From the figure, we can see multiple frequencies, multiple amplitudes, and multiple phases clearly

Fourier & Composite signals

- Composite Periodic signals → Fourier series
- Composite Aperiodic signals → Fourier transforms

My brain has too Tabs open 🤖

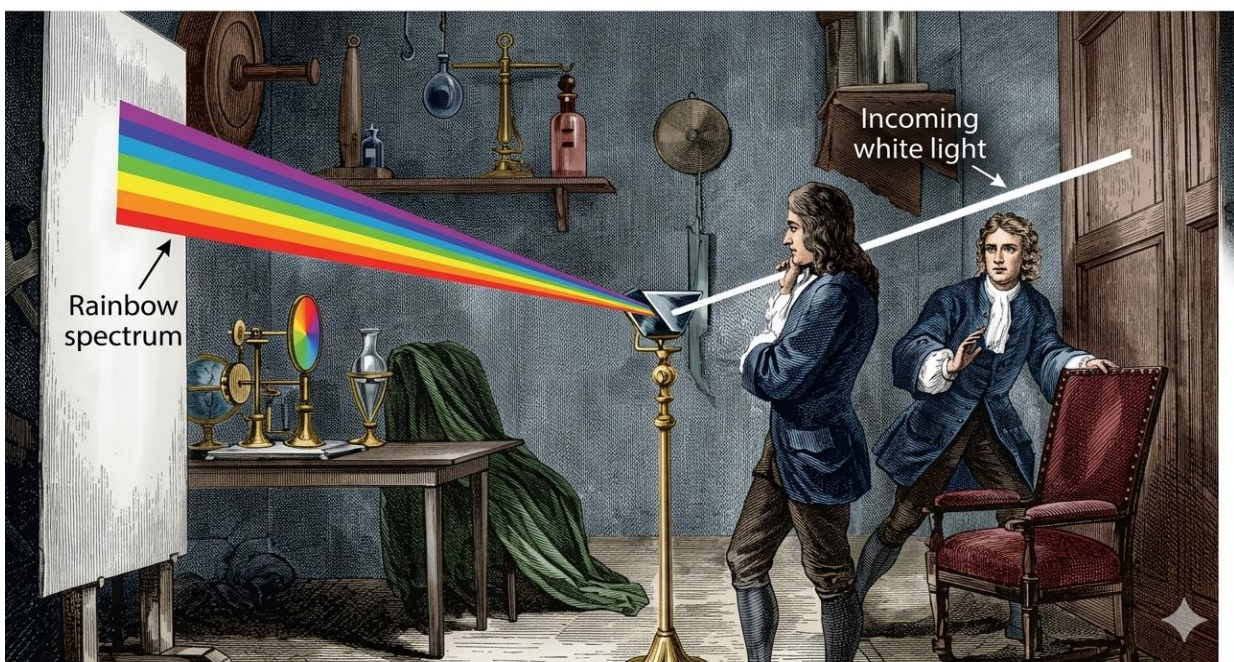
How do we study a composite signal?

A scientist named Fourier studied it many years ago. He analyzed composite signals in communications.

- First, he studied periodic signals, and the result is the Fourier Series.
- Then he studied aperiodic signals, and the result is the Fourier Transforms

What principle did Fourier rely on in his study?

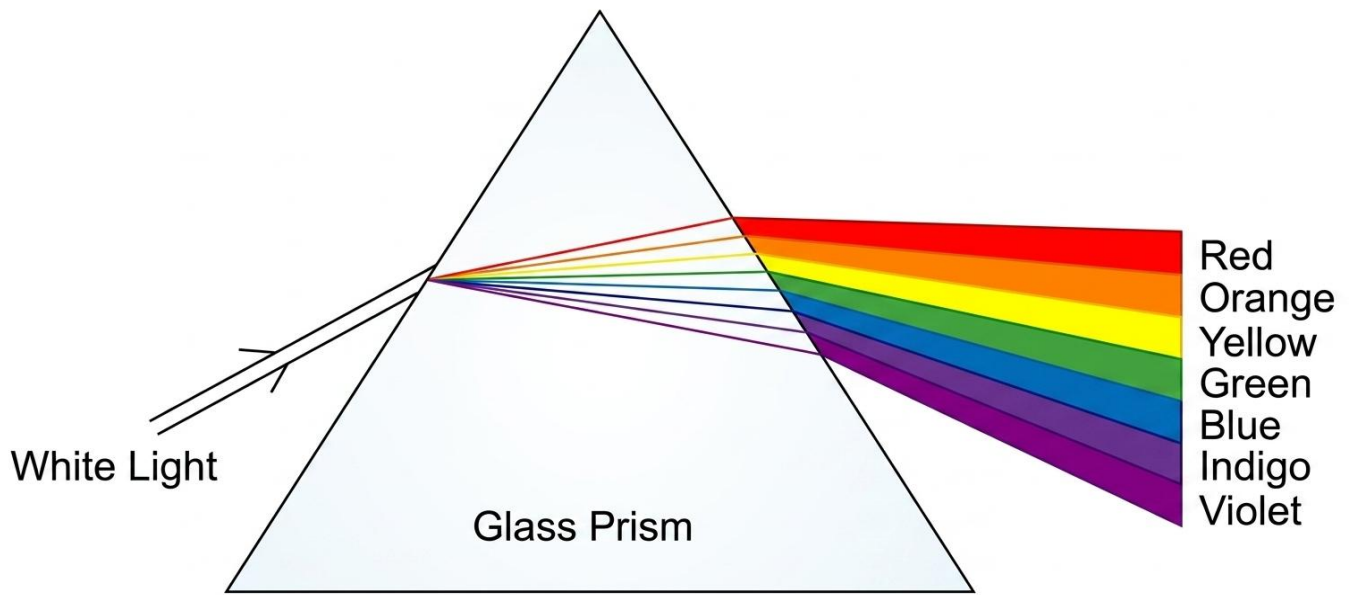
Newton's prism



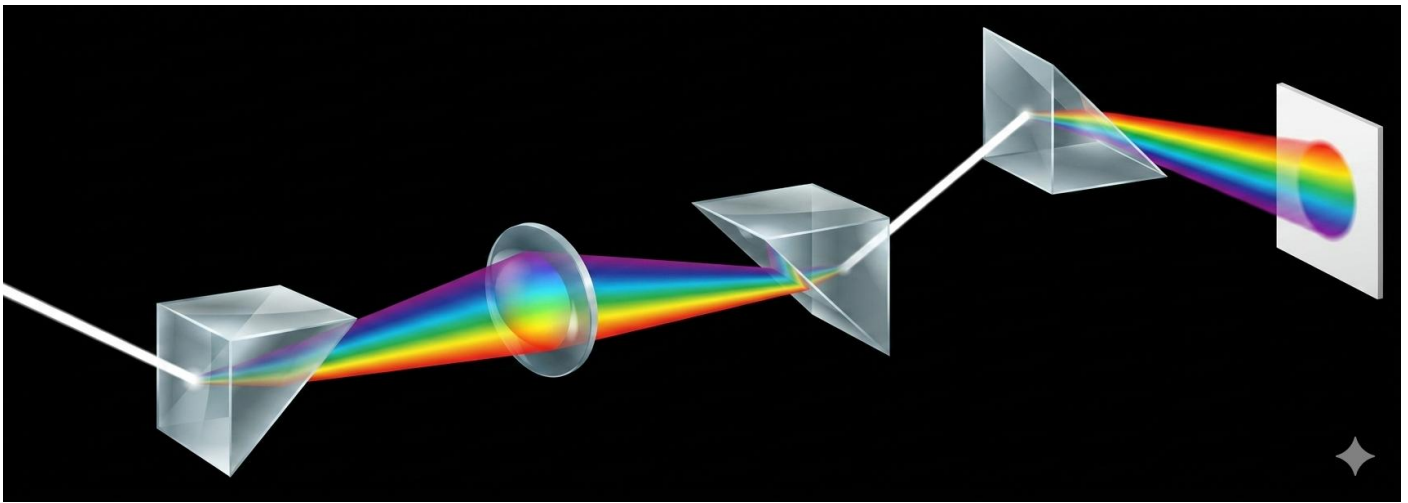
Newton's Prism Experiment

Isaac Newton conducted an experiment using a glass prism to study the nature of white light. He passed a beam of white light through a small opening and directed it toward a glass prism, then observed what happened to the light after it emerged onto a white screen

When the light passed through the prism, it was refracted and separated into a range of colors known as the **visible light spectrum**: red, orange, yellow, green, blue, indigo, and violet. This occurs because each color has a different wavelength and therefore bends at a different angle when passing through glass



The reverse Experiment



To confirm that the colors were not produced by the **prism** itself, **Newton** performed a **reverse experiment**. He collected the spectrum of colors produced by the first prism and passed them through a **second prism**.

When these colors passed through again, they did not remain separated. Instead, they were refracted in a way that reversed their previous dispersion, and they were recombined to form white light again on the screen.

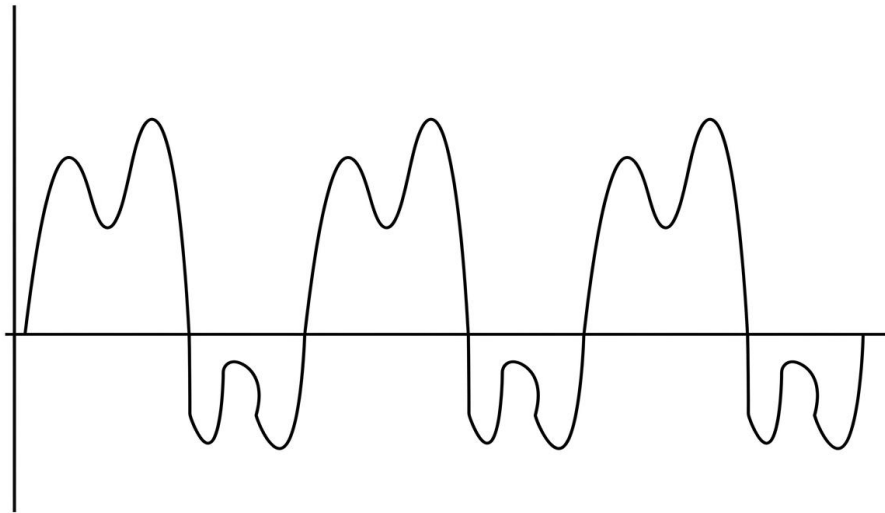
This experiment proved that white light is not a single color, but is composed of multiple colors that can be separated and recombined

Home Work:

Is there a link between Newton's experiment and the frequency spectrum from the previous lecture?

Fourier benefited from this experiment and studied the composite signal in the same way

$X(t)$ -composite, periodic signal



How to decompose it?? Using Fourier series.

First, he studied what is called a periodic signal.

From the figure, it is clear that it is a composite signal. Notice:

- 1) Multiple amplitudes
- 2) Multiple frequencies

-Why is it periodic?

Notice that in each time interval, it repeats itself in the same way

Fourier worked mathematically, while Newton worked experimentally.

For reference:

A prism is an optical transparent object used to analyze light, usually in a triangular shape with three faces.

Its main function is to refract light and separate it into its constituent colors or change its direction.

What will we work on in the composite periodic signal?

$$x(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos(nw_0 t) + B_n \sin(nw_0 t)$$

A_0 : constant (DC : Direct current)

$$w_0 = \frac{2\pi}{T_0} = 2\pi f_0$$

Why did we write $x(t)$?
Because this is a continuous composite signal.

Why did we put it from 1 to infinity? Because it is continuous.

DC: means that you have a signal with a constant part.

You either add to it to reach the maximum value, or subtract from it to reach the value you want.

This form is called the **Fourier Series**

So, the **composite signal** I have is decomposed into a constant part, a cosine part and a sine part.

This is what Fourier worked on, and we are now relying on it to solve the problem.

If we solve {1} we find:

$$x(t) = \frac{A_0}{2} + A_1 \cos(w_0 t) + A_2 \cos(2w_0 t) + A_3 \cos(3w_0 t) + \dots$$

$$+ B_1 \sin(w_0 t) + B_2 \sin(2w_0 t) + B_3 \sin(3w_0 t) + \dots$$

$A_1, A_2, A_3, \dots, B_1, B_2, B_3, \dots$ (Fourier coefficients)

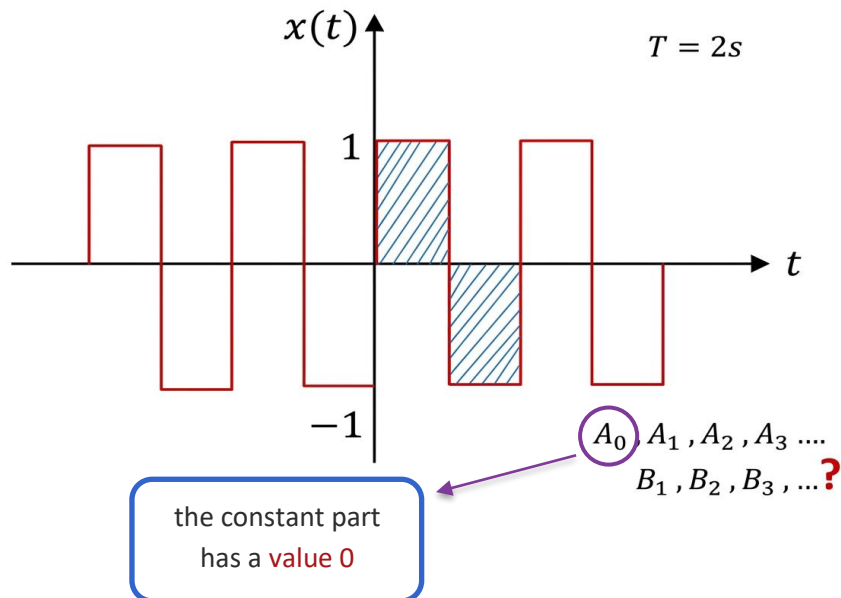
How to find these coefficients ?

$$A_n = \frac{2}{T} \cdot \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) \cos(n\omega_0 t) dt \quad ; \quad n = 0, 1, 2, \dots$$

$$B_n = \frac{2}{T} \cdot \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) \sin(n\omega_0 t) dt \quad ; \quad n = 0, 1, 2, \dots$$

Why is the interval from $-T/2$ to $T/2$? Because the signal is periodic.

Example



The signal is:

- 1) Continuous.
- 2) Periodic.

$$x(-t) = -x(t) \quad \text{Odd signal}$$

$$A_n = \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt \quad A_n = \int_0^T x(t) \cos(n\omega_0 t) dt$$

$$A_n = \int_0^1 \underbrace{x(t)}_{=1} \cos(n\omega_0 t) dt + \int_1^2 \underbrace{x(t)}_{=-1} \cos(n\omega_0 t) dt = 0$$

The result of the integration gives the same value but with opposite sign, which is why it become 0

$$\begin{aligned} B_n &= \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt \\ &= \int_0^1 x(t) \sin(n\omega_0 t) dt + \int_1^2 x(t) \sin(n\omega_0 t) dt \\ &= \int_0^1 1 \cdot \sin(n\omega_0 t) dt + \int_1^2 (-1) \sin(n\omega_0 t) dt \\ &= \left[-\frac{\cos(n\omega_0 t)}{n\omega_0} \right]_0^1 + \left[\frac{\cos(n\omega_0 t)}{n\omega_0} \right]_1^2 \\ &= \frac{-\cos(n\pi)}{n\pi} + \frac{\cos(0)}{n\pi} + \frac{\cos(2n\pi)}{n\pi} - \frac{\cos(n\pi)}{n\pi} \\ &= \frac{2}{n\pi} (1 - \cos(n\pi)) \end{aligned}$$

odd Signal : $\cos=0$ $A_n=0$

even Signal : $\sin=0$ $B_n=0$, $A_0=0$

$$x(1) = \frac{2}{\pi} (1 - \cos(\pi)) \cdot \sin(\pi t) = \frac{4}{\pi} \sin(\pi t)$$

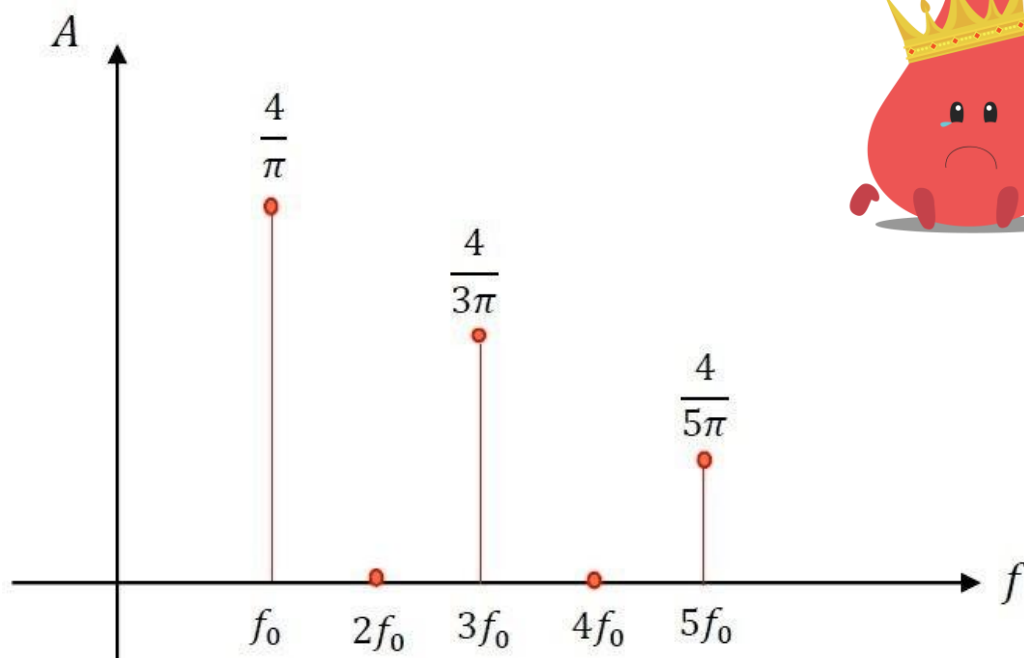
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$$x(2) = \frac{2}{2\pi} (1 - \cos(2\pi)) \cdot \underbrace{\sin(2\pi t)}_{=0} = 0$$

$$x(t) = \frac{4}{\pi} \sin(\pi t) + 0 + \frac{4}{3\pi} \sin(3\pi t) + 0 + \frac{4}{5\pi} \sin(5\pi t) + \dots$$

هذا الحد يُسمى
Fundamental
Harmonic

2nd 3rd 4th 5th



Pattern: Amplitudes decrease as frequency increases

Student Note: This shows energy is concentrated in lower frequencies. Higher harmonics contribute less to the overall signal shape.

Key Takeaways

- **Domain choice matters:** Use time domain for simple signals; frequency domain for composite signals
- **Bandwidth matching is critical:** Medium must support signal bandwidth to avoid data loss
- **Fourier bridges domains:** Converts between time (what we measure) and frequency (what we analyze)
- **Signal symmetry simplifies math:** Odd signals → only sine terms; Even signals → only cosine terms
- **Real signals need finite approximation:** Although formulas show infinity, we use finite terms in practice

“THE END”

